Homework 1: Exploring Astah and its Features

Start Assignment

- Due Saturday by 11:59pm
- Points 60
- Submitting a text entry box, a website url, or a file upload
- File Types pdf
- Available Jan 13 at 12am - Jan 24 at 11:59pm

Objective

To become familiar with advanced features of Astah and apply object-oriented design principles using UML diagrams.

Prerequisite:

You need to have the Astah student version downloaded and Installed.

https://astah.net/products/free-student-license/ (https://astah.net/products/free-student-license/)

Problem 1 [20 pts] - Astah Setup and Basic Class Diagram

1. Download and install Astah Professional (or the highest version available to students). You can find installation guides at Astah User Guide.
2. Go to "Tools" -> "System Properties" -> "Project" and set the default programming language as "Java".
3. Create a new project and a class diagram in Astah.
4. Create a class named "Student" with the following:
   - Public attributes:
     - "name" (String)
     - "studentID" (String)
   - Private attributes:
     - "gpa" (double)
     - "enrolledCourses" (ArrayList<String>)
   - Public methods:
     - "enrollInCourse" (void)
     - "calculateGPA" (double)
     - "dropCourse" (void)
5. Create another class named "Course" with the following:
   - Public attributes:
     - "courseID" (String)

    - "courseName" (String)
  - Private attributes:
    - "instructor" (String)
    - "maxCapacity" (int)
    - "enrolledStudents" (ArrayList<String>)
  - Public methods:
    - "addStudent" (boolean)
    - "removeStudent" (boolean)
    - "getAvailableSeats" (int)
6. Establish an association relationship between the Student class and the Course class. This should be a many-to-many relationship, as a student can enroll in multiple courses and a course can have multiple students.
7. Submit a screenshot of your class diagram.

Problem 2 [20 pts] - Advanced Class Relationships

A. Expand your class diagram:
   1. Create an abstract class named "Person" with the following:
      - Protected attributes:
        - "name" (String)
        - "age" (int)
      - Abstract method:
        - "getDescription" (String)
   2. Create an interface named "Gradable" with the following method:
        - "calculateGrade" (double)
   3. Show inheritance by making the "Student" class inherit from the "Person" class.
   4. Show implementation by making the "Student" class implement the "Gradable" interface.
   5. Create a new class named "Department" with the following:
      - Public attributes:
        - "departmentName" (String)
        - "departmentCode" (String)
      - Private attribute:
        - "courses" (ArrayList<Course>)
      - Public methods:
        - "addCourse" (void)
        - "removeCourse" (void)
   6. Show composition by creating a composition relationship between "Department" and "Course". This indicates that courses are an integral part of a department and cannot exist without it.
   7. Submit a screenshot of your class diagram.
B. To generate Java code:

1. Go to "Tools" -> "Java" -> "Export Java"
2. Choose your export settings and location

Upload the **Astah generate** code into your google drive the share the link. Make sure you give the permission so that we can download the code. Points will be given only to the code exported from Astah.

Problem 3 [20 pts] - Astah Plugins

1. To install a plugin:
   - Go to the Project bar main menu -> "Plugin" -> "Plugins Site.."
   - Choose a plugin and follow the installation instructions
2. Use the plugin according to its specific instructions.
3. Write your review in a document.
4. Submit proof of your plugin usage (screenshot or exported file) and your written review.

Note 1: For all diagram submissions, ensure that your diagrams follow UML standards and best practices for clarity and readability. You can refer to **Astah's documentation (https://astah.net/support/astah-pro/user-guide/)** for more detailed instructions on using specific features.

Note 2: Since this homework involves a lot of subjectivity it will be graded based on your understanding of software design principles and your ability to use tools like Astah effectively. Don't hesitate to be creative and explain your thought process in your submissions.

Submission

1. Review each Problem's submission instructions and use the HW1 submission template given **CSE 460 Homework - 1_ Exploring Advanced Astah Features_Submission Template.docx (https://canvas.asu.edu/courses/248379/files/123147504?wrap=1)**
2. Save your file as *LastnameFirstInitial_HW#1* (Ex. SmithR_HW#1)
3. On the top of this page select Start Assignment
4. In the 'File Upload' section, select Upload File
5. Use the Choose File SmithR_HW#1.docx option to locate and upload your saved file.
6. Be sure to hit Submit Assignment to finish submission

Some Rubric

Criteria	Ratings		Pts
Astah Setup	**2 pts** **Full Marks**	**0 pts** **No Marks**	2 pts
Class Diagram Structure	**12 pts** **Full Marks**	**0 pts** **No Marks**	12 pts
Relationships	**6 pts** **Full Marks**	**0 pts** **No Marks**	6 pts
Person Class	**4 pts** **Full Marks**	**0 pts** **No Marks**	4 pts
Gradable Interface	**4 pts** **Full Marks**	**0 pts** **No Marks**	4 pts
Inheritance and Implementation	**6 pts** **Full Marks**	**0 pts** **No Marks**	6 pts
Department Class and Composition	**6 pts** **Full Marks**	**0 pts** **No Marks**	6 pts
Plugin Installation	**4 pts** **Full Marks**	**0 pts** **No Marks**	4 pts
Plugin Usage	**8 pts** **Full Marks**	**0 pts** **No Marks**	8 pts
Written Review	**8 pts** **Full Marks**	**0 pts** **No Marks**	8 pts
			Total Points: 60